BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions)

Maximum Marks: 50

Note: This section consists of 25 part questions and all are to be answered. Each question carries 02 marks. Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part. iii) The use of calculator is allowed. All notations are used in their usual meanings. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 2 Hours — Maximum Marks: 100 (As per condensed syllabus)

Q1. Choose the correct answer for each from the given options:

Qi. Let A = {0,1}, B = {1,2}, C = {2,3}, then A x (B ∩ C):

(A) φ
(B) {(1,3), (0,1)}
(C) {(0,2), (1,2)}
(D) {(2,3), (1:1)}

Qii. If A and B are two non-empty sets, then A ∩ (B ∪ φ) is equal to:

(A) A
(B) B
(C) A∩B
(D) φ

Qiii. The real and imaginary parts of (√5+i)/(√5−i) are:

(A) −2/3 and √5/3
(B) 2/3 and √5/3
(C) −2/3 and −√5/3
(D) 2/3 and √5/3

Qiv. The multiplicative inverse of (a, b) is:

(A) $(-a/(a^2+b^2), -b/(a^2+b^2))$
(B) $(a/(a^2+b^2), -b/(a^2+b^2))$
(C) $(-a/(a^2+b^2), b/(a^2+b^2))$
(D) $(a/(a^2+b^2), b/(a^2+b^2))$

Qv. If $Z = 3i + 2$, then $Z + \bar{Z}$:

(A) 6i (B) 6

(C) 0 (D) 4

Qvi. If the product of the roots of equation $px^2 - 4x + 4 = 0$ is 2, then the value of p is:

(A) 1 (B) 2

(C) 4 (D) −4

Qvii. The value of $i^{57} + 1/i^{125}$ is:

(A) 0 (B) 2i

(C) −2i

Qviii. If $\omega^3 = 1$ and $\omega \neq 1$ then $(1+\omega)(1+\omega^2)(1+\omega^3)\ldots\ldots\ldots = :$

(A) 3 (B) −3

(C) 1 (D) −1

Qix. If the sum of the roots of $6x^2 - 3mx + 5 = 0$ is equal to the product of its roots, then the value of m is:

(A) 3/3 (B) 3/5

(C) 5/3 (D) −3/5

Qx. The next term of the sequence 1,2,4,7,11, ... is:

(A) 15 (B) 16

(C) 17 (D) 18

Qxi. The 15th term of an A.P. is -15, -22, -29, -36, ... is:

(A) 106 (B) −106

(C) 113 (D) −113

Qxii. $1, x^2, 6 - x^2$ will form a G.P if x = :

(A) 2 (B) √2

(C) 4 (D) 8

Qxiii. The H.M between 1/3 and 2/5 is:

(A) 11/4 (B) 4/11

(C) −4/11 (D) −11/4

Qxiv. The number of ways of sitting n persons at a round table is:

(A) (n−1)! (B) (n+1)!

(C) 1/2(n+1)! (D) 1/(n−1)!

Qxv. The middle term in the expansion of $(1+x)^9$ is:

(A) 6 (B) 5

(C) 4 (D) 3

Qxvi. If $(a+b)^n$, where n is an even number then the number of middle term/terms are:

(A) One (B) Two

(C) Three (D) Four

Qxvii. If $(a-b)^{13}$, then it will contain:

(A) 14 terms (B) 13 terms

(C) 15 terms (D) 16 terms

Qxviii. The central angle of an arc of a circle whose length is equal to the radius of the circle is called:

(A) radian (B) degree

(C) minute (D) second

Qxix. In ΔABC, $\cos\alpha/a = \cos\beta/b = \cos\gamma/c$, then ΔABC is a/an:

(A) equilateral triangle (B) Isosceles triangle

(C) right angled triangle (D) obtuse angled

Qxx. $\cos(\pi/2)\cos(\pi/6) - \sin(\pi/2)\sin(\pi/6)$ is equal to:

(A) 1/2 (B) $-\sqrt{3}/2$

(C) $\sqrt{3}/2$ (D) 1/2

Qxxi. $\cos 2\theta$ =:

(A) $1+2\sin^2\theta$ (B) $2\cos\theta-1$

(C) $\cos^2\theta+\sin^2\theta$ (D) $2\sin\theta\cos\theta$

Qxxii. If $\sin\theta > 0$ and $\cos\theta < 0$, then $p(\theta)$ lies in this quadrant:

(A) 1st (B) 2nd

(C) 3rd (D) 4th

Qxxiii. In a triangle ABC, $\sin(\alpha/2)$ =:

(A) $\sqrt{((s-b)(s-c)/bc)}$ (B) $\sqrt{((s-a)(s-b)/ab)}$

(C) $\sqrt{((s-a)(s-c)/ac)}$ (D) $\Delta/(s-a)$

Qxxiv. $\cos(270+\theta)$ =:

(A) $\cos\theta$ (B) $\tan\theta$

(C) $\sin\theta$ (D) $\sec\theta$

Qxxv. The Principal value of arc sin $(-\sqrt{3}/2)$:

(A) $\pi/3$ (B) $\pi/6$

(C) $-\pi/3$ (D) $-\pi/6$

Section 'B' (Short-Answer Questions)

Maximum Marks: 30

Note: Attempt any Six part questions from this section, Selecting at least Two part questions from each equations. All questions carry equal marks.

Q2. COMPLEX NUMBER, ALGEBRA

(i) Solve the complex equation (x, y) (2, 3) = (-4, 7)

(i-OR) Solve the complex equation for x and y: $x(1 + 2i) + y(3 + 5i) = -3i$

(ii) Show that $Z = 1 + i$ and $Z = 1 - i$ satisfy the equation $Z^2 - 2Z + 2 = 0$

(iii) Find all the cube roots of 27 in terms of 'ω'.

(iv) If α, β are roots of the $y^2 - qy + p = 0$, form an equation whose roots are doubled the roots of the given equation.

(iv-OR) If α, β are roots of equation $2x^2 + 3x + 4 = 0$, form an equation whose roots are $1/\alpha$ and $1/\beta$.

Q3.

(i) If ${}^{n}P_4 = 24 \cdot {}^{n}C_5$, Find n.

(ii) How many terms of the series 10 + 8 + 6 + ... will make the sum 24?

(ii-OR) What term of the sequence 3, 5, 9, 15, ... is 75.

(iii) If nth term of the sequence 12, 24, ... is the same as that of 256, 128, 64, ... Find n.

(iv) Prove by Mathematical Induction $1 + 3 + 5 + ... + (2n - 1) =$ [illegible in source scan] where n is positive integer.

(iv-OR) Prove that $3^{3n} - 26n - 1$ is divisible by 676 where n is any positive integer.

Q4. TRIGONOMETRY

(i) A belt 24.75cm long passes around a 3.5cm diameter pully. As the belt makes three complete revolutions in a minute. How many radians does the wheel turn in one second?

(i-OR) How far does a boy on a bicycle travel in 15 revolutions if the diameter of the wheel of his bicycle each equal to 56cm.

(ii) Prove that $\sin^2(\pi/3) + \cos^2(\pi/3) = \sin^2(\pi/6) - \cos^2(\pi/6)$ without using calculator.

(ii-OR) Find the value of $\sin(7\pi/12)$ without using calculator

(iii) For any ΔABC, Drive law of cosine.

(iii-OR) Solve the triangle ABC in which a = 5cm, b = 10cm and c = [illegible in source scan]

(iv) Solve: $2\sin^2 x + 2\sqrt{2}\sin x - 3 = 0$

(iv-OR) $\cos\theta + \cos 2\theta + 1 = 0$

Section 'C' (Detailed Answer Questions)

Maximum Marks: 20

> Note: Attempt Two questions from this section.

Q5.

(a) Insert four Harmonic Means between 12 and 48/5.

(b) Find the term independent of x in $(x - 2/x)^{10}$

(b-OR) Find the middle term in the expansion of $(a/y - \sqrt{y})^{16}$

Q6.

(a) The measure of the two sides of a triangle are 4 and 5 units. Find the third side so that the area is 6 square units.

(b-OR) Solve the system of equations: $4x + 3y = 25$, $4/x + 3/y = 2$

(b-OR) Solve the equation $\sqrt{(2x+7)} + \sqrt{(x+3)} = 1$

Q7.

(a) The three sides of triangular building have 10cm, 11cm and 13cm respectively. Find the measure of its largest angle and also the area of the building.

(a-OR) Does the expansion $(x + 1/x)^7$ contains a term involving x^6 ?

(b) Prove any two of the following: (i) $\sin 2\theta/\sin\theta - \cos 2\theta/\cos\theta = \sec\theta$ (ii) $(1 + \tan^2(\pi/6))/\tan^2(\pi/6) = \operatorname{cosec}^2(\pi/6)$ (iii) $\sqrt{((1-\sin\theta)/(1+\sin\theta))} = \sec\theta - \tan\theta$ (iv) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

> نوٹ: سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔